

Sandbox Practice Question

In this sandbox-style question, you must use the information and tools you are given, combined with your basic feedback control knowledge, to describe how you would solve a more complete and open-ended problem through the stages of modelling, analysis and controller design.

You are provided with a template to help you structure your answer, and give you hints of things to think about in each section. Just describe what you would do and why, supported by whatever calculations and visualizations are needed, and you'll get marks for taking correct steps, and demonstrating your understanding. Follow-through marking will be applied, so you will still get marks for doing the right thing from where you are, even if something went wrong in the previous step.

There are 55 marks available, with a mark allocation of 45.

Problem Statement:

At some point in the future, UCT has decided to reinstate the pub at the sports centre. There is a problem, though: after a big lunch, accompanied by as many beers as the day's struggles demand, nobody feels like walking back up all the stairs to Menzies, so many productive afternoons of engineering education are being lost! The Mechanical Engineering Department proposes a solution: a ski lift to carry people back up to Plaza. The African Robotics Unit (ARU) worries that intoxicated, overfed passengers plus swaying in the Cape Town wind is a very messy equation. They decide the lift needs a reaction wheel to keep it stable. You are the intern tasked with designing the feedback controller for this system.

Information:

Method of operation:

Passengers will hang beneath the cable in individual pods. The angle of the pod with respect to vertical is θ , and ideally, we want this to be as close to zero as possible. Each pod P is assumed to have a mass moment of inertia J of around 600 kg/m^2 . The equation of motion for the swinging pod is

$$J\ddot{\theta}(t) = \tau(t) - b\dot{\theta}(s)$$

where τ is the torque exerted on the pod, and b is a damping coefficient quantifying the mechanical resistance to swinging in the system.

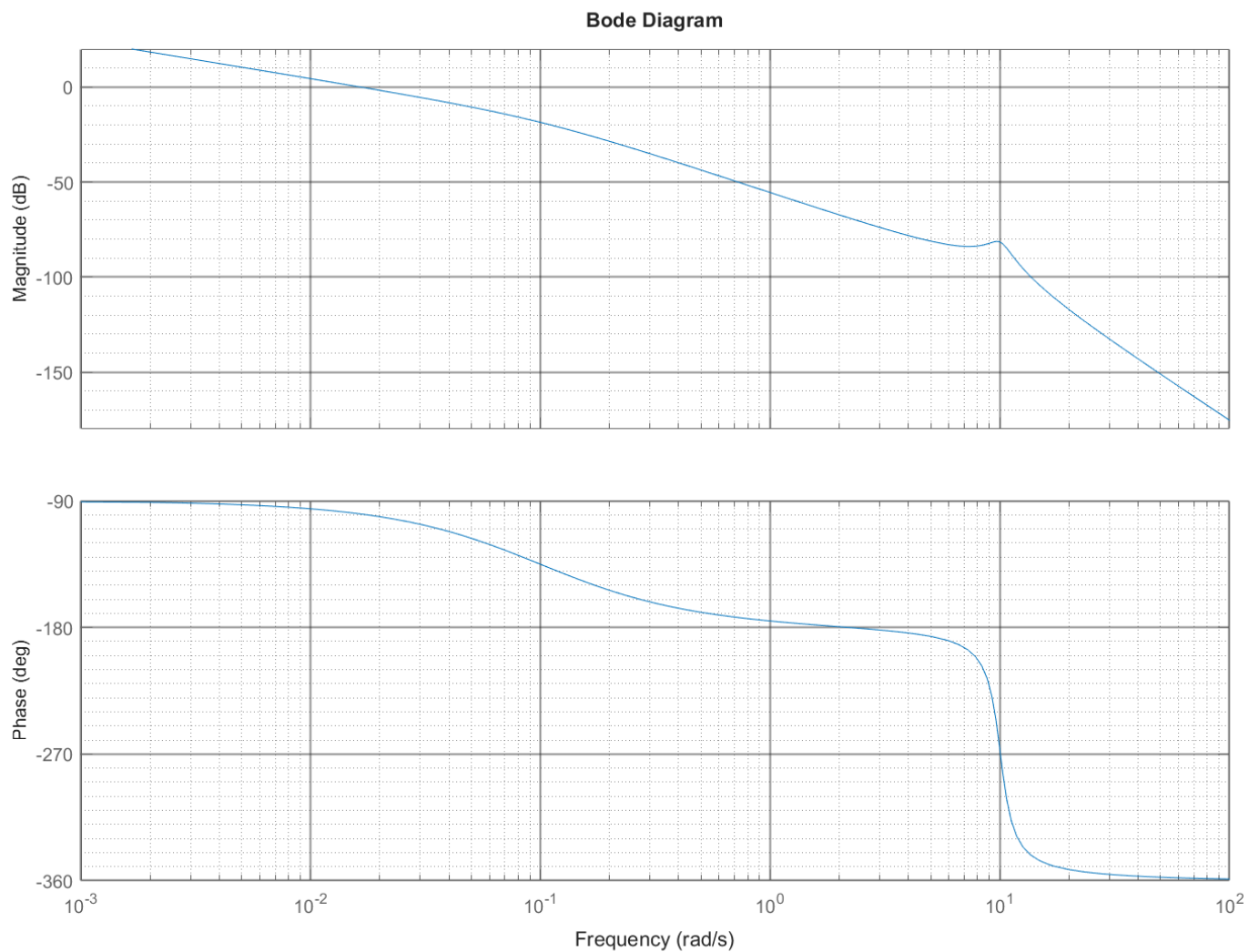
The torque is produced by spinning the wheel with a motor, M . The torque required to move the wheel produces an equal and opposite reaction torque acting on the pod. In this extremely simplified view of this problem, you do not need to consider the motion of the wheel – rather, we can think of the motor as applying the reaction torque directly to the pod.

An inertial measurement unit $H(s)$ feeds back the angle of the pod to the microcontroller $C(s)$ driving the motor. This sensor is sufficiently accurate that you may work under the assumption of unity feedback.

Pod dynamics:

It is possible that, in some wind conditions, turbulent air flow around the pod could produce vibrations in the system. To test the effects on the pod, the mechanical engineers perform a simulation subjecting it to torque disturbances at different frequencies. The results of this test (shown in the following Bode plot) can be used to determine the mechanical parameters of the system and also to identify frequencies where potentially unmodelled elements could cause resonant behavior. Care should be taken to ensure that the closed-loop system does not excessively magnify disturbances at these frequencies.

Under most operating conditions, the ski lift is unlikely to be subjected to vibrations at frequencies above 1 rad/s.



Motor:

The motor is a powerful DC motor attached to a gearbox. The gearbox scales the torque produced by the motor up by a factor of 10. The torque produced by the motor is directly proportional to the current across it, and related to the voltage e applied to it by the transfer function:

$$M(s) = \frac{10K_t}{Ls + R}$$

where L and R are the inductance and resistance of the motor coil, and K_t is the motor torque constant.

Estimates of these values from the datasheet are given in Table 1, but you wish to validate these parameters through system identification.

Table 1: Motor Parameters	
R	20 Ω
L	4 mH
K_t	100

To obtain the electrical parameters, you perform a locked-rotor step test, using a voltage step of magnitude 12V, which produces the current response shown below.

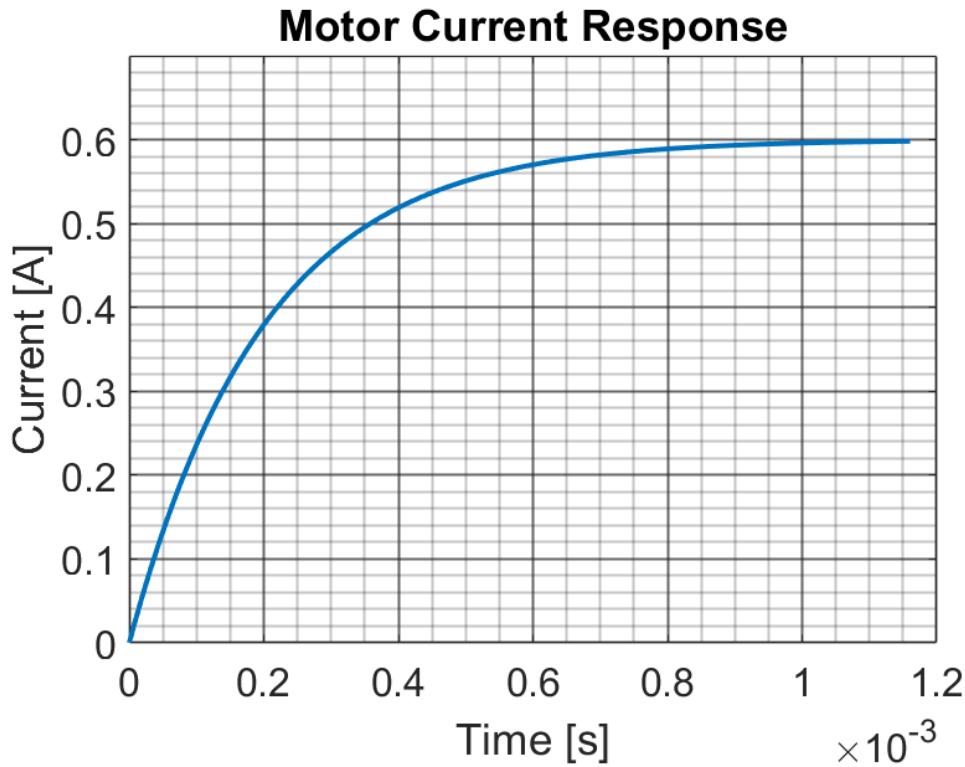


Figure 1: Motor current in response to 12V voltage step

(Note: the transfer function relating voltage and current i is $\frac{i(s)}{e(s)} = \frac{1}{Ls+R}$.)

To confirm the torque constant, you use the motor to spin the wheel. The wheel velocity $\dot{\theta}_t$ is related to the input velocity by the transfer function

$$\frac{\dot{\theta}_t}{e} = \frac{10K_t}{(Ls + R)(J_t s + b_m)}$$

where J_t is the mass moment of inertia of the wheel, and b_m is a parameter representing damping in the motor and gearbox. Based on the wheel's geometry, you estimate J_t to be around 6 kg/m².

The wheel velocity response to a 12V step is shown below.

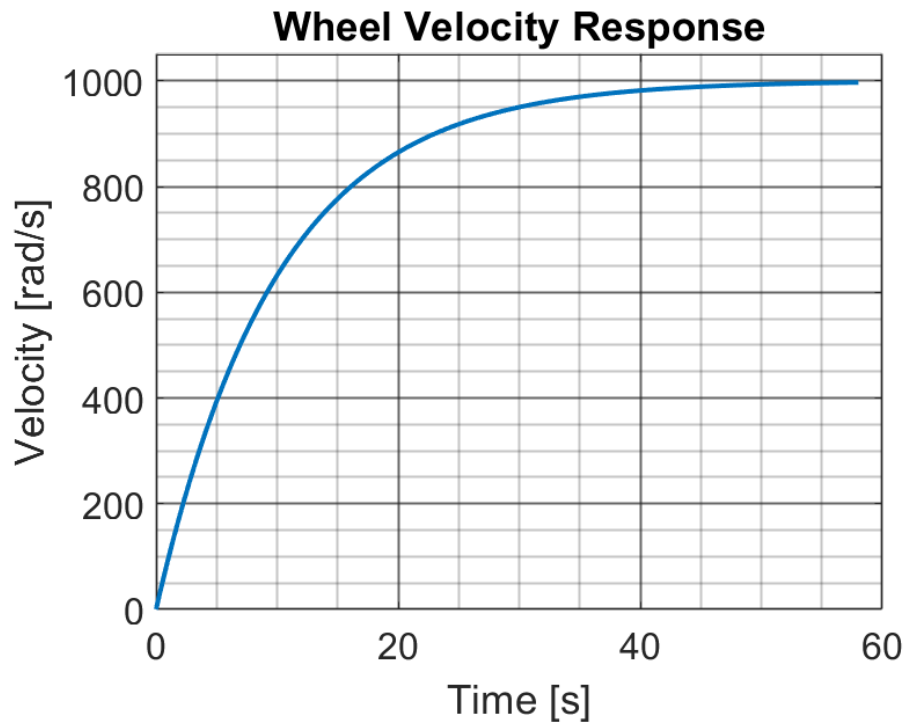


Figure 2: Reaction wheel velocity in response to 12V input step applied to motor

Specifications

Prepare two controller designs, for different design priorities

- 1) Ride comfort:
 - No overshoot
 - No limitation on settling time
- 2) Fast correction:
 - Low overshoot ($< 10\%$)
 - Settling time to within 2% of the final value in less than 8 seconds.

Both controllers should correct the position with no error.

BONUS: Comment on whether the design will be sensitive to disturbances caused by resonance in the mechanical system i.e. cause these to increase by more than a factor of 2. (You don't need to design around this, just comment)

You may use any combination of proportional, integral, lead and lag stages to design your controllers. Comment on whether your design is realistic (you don't have to redesign if it isn't – just give a recommendation for how this could be achieved).

If you cannot model the system, you may use the open-loop transfer function $G(s) = \frac{0.833}{s(10s+1)}$

Template

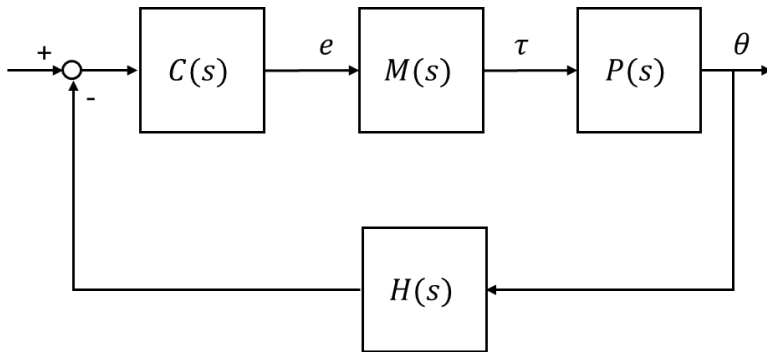
You may structure your answer using the following sections:

- 1) Modelling (5 marks)
block diagram model, open loop transfer function in terms of the unknown parameters of the system
- 2) System identification (15 marks)
evaluation of the unknown parameters from the provided experimental results, and simplification of the model where applicable
- 3) Controller design 1 (10 marks)
Specification analysis, controller selection and design
- 4) Controller design 2 (25 marks)
Specification analysis, controller selection and design

Solution

Modelling [5 marks]:

The block diagram of the system is as follows:



[3 marks]

The open-loop transfer function is therefore $G(s) = M(s)P(s)$. [1 mark]

The form of $P(s)$ can be obtained by taking the Laplace transform of the given equation of motion, assuming initial rest:

$$P(s) = \frac{1}{s(Js+b)} \text{ [1 mark]}$$

Combining with $M(s)$ gives $G(s) = \frac{10K_t}{s(Ls+R)(Js+b)}$ [1 mark]

System Identification [15 marks]:

The only known parameter given is the mass moment of inertia. The other parameters must be evaluated from the given frequency and step responses.

Pod Dynamics:

Due to the integrator, the DC gain is effectively infinite, but the dominant pole $\frac{b}{J}$ can be determined by finding the frequency where the phase of the response is -135 degrees. (The pole at 0 reduces the phase at all frequencies by 90 degrees, so this is effectively the same as finding the crossing at -45 degrees.)

This occurs at around 0.1 rad/s, indicating that $b = 0.1(600) = 60$. [2 marks]

There is also a resonant peak at around 10 rad/s, indicating a complex pole pair at this frequency, but since it is outside the typical operating frequency range and much faster than the dominant poles, we can exclude it from the model. [1 mark]

However, we should note that frequencies in the vicinity of 10 rad/s must be attenuated by the final closed-loop system (see: specification analysis.)

Motor:

For the electrical properties of the motor L and R , we will use the transfer function $\frac{1/R}{s+1}$, applied to the motor current step response. [2 marks for any attempt using correct response and mapping of parameters]

The final value of the response step is 0.6, giving a gain of $\frac{1}{R} = \frac{0.6}{12} = 0.05$ when divided by the magnitude of the input step (12V). This gives $R = \frac{1}{0.05} = 20\Omega$. [1 mark]

The time constant of the response, $\frac{L}{R}$, is given by the time when the step reaches around 63% of its final value i.e. 0.38, which occurs at 0.2×10^{-3} seconds. This gives $L = 4 \times 10^{-3}$ H. [1 mark]

These values are consistent with those in the datasheet. [1 mark]

The tail velocity response can be used to find the remaining unknown, K_t .

By inspection of the tail velocity response, the time constant of this response is much longer than that of the current response, so the pole of the electrical system can be ignored, and the tail velocity transfer function simplified to:

$$\frac{\frac{10K_t}{R}}{J_t s + b_m} = \frac{\frac{10K_t}{R b_m}}{\frac{J_t}{b_m} s + 1} = \frac{\frac{0.5K_t}{b_m}}{\frac{6}{b_m} s + 1}$$

[2 marks for any attempt using the correct response and mapping of parameters]

The final value of the step is $12 \frac{0.5K_t}{b_m} = 1000$. [1 mark]

It reaches 63% of this value at the time constant, $\frac{6}{b_m} = 10$ seconds. [1 mark]

This gives $b_m = 0.6$, and hence, $K_t = 100$, which also agrees with the datasheet. [1 mark]

Final Open-Loop Model:

The mechanical components of the system have a much longer time constant than the electrical ones do, so the order of $G(s)$ can be reduced by replacing them with the gain $\frac{1}{R}$. This gives

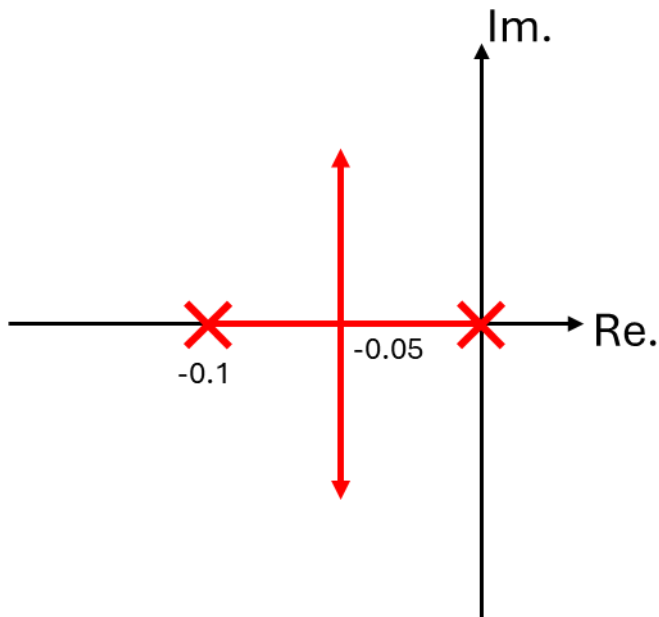
$$G(s) = \frac{\frac{10K_t}{R}}{s(Js + b)} = \frac{50}{s(600s + 60)} = \frac{0.833}{s(10s + 1)}$$

[2 marks]

Controller Design 1 [10 marks]:

Specification Analysis

For the system to have no overshoot, the poles must be real, i.e. the system must be critically- or overdamped. [1 mark]



A root locus for the system is shown above. [2 marks]

Since there are real pole locations on this root locus, we can potentially use a proportional controller to satisfy this set of specifications. [2 marks]

The system has one pole at the origin, so it is type 1 and will track with 100% accuracy regardless of the gain. [1 mark]

Design

The system will be critically damped when the poles are both at -0.05. [1 mark]

This occurs at a gain of $K = \frac{1}{|G(-0.05)|} = 0.03$. [1 mark]

You don't need an inverse Nichols chart to work with the sensitivity! We just need to calculate the magnitude of $\frac{1}{1+KG(j\omega)}$ at the frequency of interest i.e. 10 rad/s. We find $|\frac{1}{1+0.03G(j10)}| = 1$, which means that these disturbances will not be magnified. [2 marks]

Controller Design 2 [25 marks]:

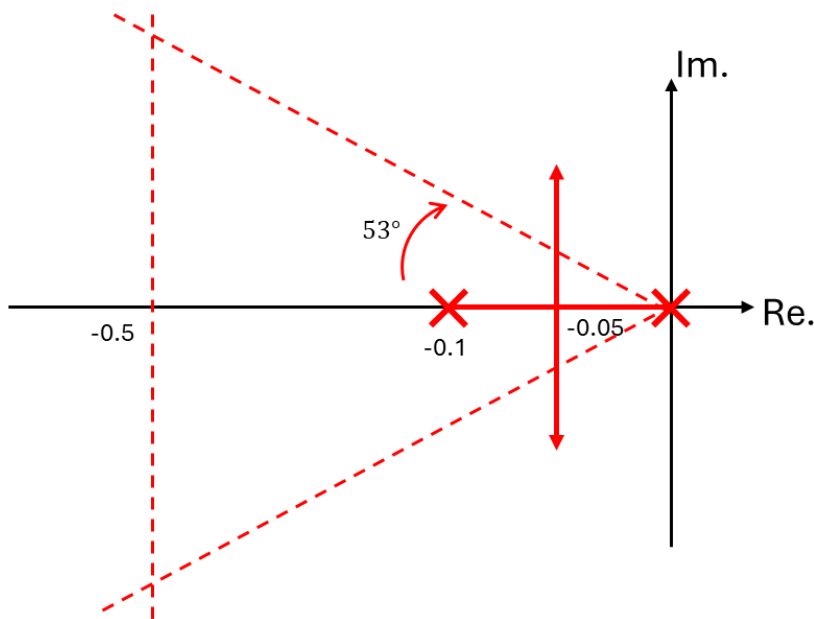
Specification Analysis:

From the formula for %OS, less than 10% overshoot corresponds to a damping ratio of more than 0.6. [1 mark]

This means the poles must be within an angle of $\cos(0.6) = 53^\circ$ with respect to the negative real axis.

The 2% settling time is around 4 times the time constant, so settling in 8 seconds would indicate a time constant of 2 and hence, poles left of -0.5. [2 marks]

This leads to the specification region below on the root locus: [2 marks]



There are no locations where both poles are inside the specification region on the root locus, so a lead compensator is required for this design. [2 marks]

(We can only calculate the maximum gain for the sensitivity requirement once we've designed the compensator stage.)

Design:

Any target pole pair within the specification region: [1 mark]

For this solution, we will use $-0.5 + 0.5j$

The angles of the existing system poles to the desired pole $s_d = -0.5 + 0.5j$ are:

135 degrees for the pole at 0, [1 mark]

and 128.7 degrees for the pole at 0.1 [1 mark]

This gives a total angular contribution of -263.7 for both. [1 mark]

The compensator must therefore add 83.7 degrees. [1 mark]

The zero will contribute this angle alone when it is at $-0.5 - \frac{0.5}{\tan(83.7^\circ)} = -1$ [1 mark].

We also want the zero to be left of the desired pole, so this gives a range between -0.55 and -0.5 for the zero. [1 mark]

Any zero location within the range = [1 mark]

Because the system is already type 1, we don't need to worry about getting enough gain with the zero, so we can play it safe and choose something close to the upper limit to ensure a feasible controller.

Let's try putting it directly below the target pole at -0.5, for an angle of 90 degrees. [1 mark]

The pole must then add 6.3 degrees [1 mark]

It will do this when it is positioned at $-0.5 - \frac{0.5}{\tan(6.3^\circ)} = -5$. [1 mark]

This is less than 10 times further from the origin than the zero is, so the compensator is physically realizable. [3 marks for any sensible conclusion and recommendation in the case of it being not realizable.]

The compensator is then $C(s) = \frac{2s+1}{0.2s+1}$ [1 mark]

The gain at which the desired pole will be obtained is $K = \frac{1}{C(s_d)G(s_d)} = 4.92$ [1 mark]

At this gain, the sensitivity at 10 rad/s will be $\frac{1}{|1+4.92C(j10)G(j10)|} = 1.03$, so this controller also won't magnify the resonant disturbances excessively. [2 marks]